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RA-11: Superset-specific controls for Kronecker trace estimation

A smaller exponential factor, finite-bit nonadaptive queries,
and an exact theorem for real-separable binary factors

Prepared with ChatGPT; proofs, exact arithmetic checks, and reproducible diagnostics

Scope: partial results, not a complete solution of RA-11. This note improves the all-PSD upper bound from the preceding package and proves an exact-query theorem for a larger promised class. It does not match the general lower bound as the tensor order grows. The arguments and implementations have been internally checked, but have not received independent mathematical review or formal proof-assistant verification.

Abstract

Let an arbitrary real PSD matrix of order n^q be accessible through full matrix-vector responses to real product inputs. We construct a nonadaptive, finite-random-bit estimator using at most

$$\min \left\{ n^q, \left\lceil \frac{2^{73} q(q+1) \Gamma_n^q}{\varepsilon} \right\rceil \right\}, \quad \Gamma_n^2 = \frac{(7n+4)(53n+12)}{48(n+1)^2},$$

original oracle calls, with relative-error success probability at least $3/4$. In particular, $\Gamma_n < \sqrt{371/48} < 2.78014$. The new step is a separate control for each partial trace: its contraction noise depends only on errors in strict supersets. A finite-support probe distribution has exactly the complex-spherical second and fourth moments; integer sketch weights remove all square roots from the implementation. Geometric error budgets and zero approximations at inactive stages give the displayed bound for the entire accuracy range. Separately, for $n = 2$, exact trace estimation of all real-separable PSD matrices requires and suffices to use $q + 1$ queries, even with randomized adaptive algorithms. The same exact rate holds for the larger cone of PSD matrices invariant under every local transpose. The unresolved general dependence on q is stated explicitly.

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1 Target, new results, and the unresolved requirement

1.1 The oracle and the requested minimax quantity

Fix integers $n, q \geq 2$, set $N = n^q$, and let $0 < \varepsilon < 1/2$. The unknown input is any real symmetric matrix $M \succeq 0$ of order N . An oracle call specifies $v_1, \dots, v_q \in \mathbb{R}^n$ and returns the entire vector

$$M(v_1 \otimes \dots \otimes v_q).$$

The quantity $Q(n, q, \varepsilon)$ is the smallest worst-case call count of a randomized adaptive algorithm such that, for every fixed $M \succeq 0$,

$$\mathbb{P}\{|\hat{\tau} - \text{tr } M| \leq \varepsilon \text{tr } M\} \geq 2/3. \quad (1)$$

Arithmetic cost is not charged; finite exact computation is permitted, and no conditioning restriction is imposed. These are the requirements of the repository statement [1]. The matrix itself is not assumed to be a tensor product. Write Q_{na} for the stronger requirement that all query inputs be selected before observing any responses.

1.2 The new upper bound

Throughout the upper-bound construction, put

$$\eta = \frac{n}{n+1}, \quad \beta = 2\eta, \quad b = 1 + \frac{3\eta}{4}, \quad a = 1 + \frac{41\eta}{12}, \quad \Gamma_n = \sqrt{ab}, \quad K = 2^q - 1. \quad (2)$$

Thus

$$\Gamma_n^2 = \frac{(7n+4)(53n+12)}{48(n+1)^2}, \quad \sup_{n \geq 2} \Gamma_n = \sqrt{371/48} = 2.78013788 \dots$$

Theorem 1.1 (Superset-control upper bound). *For every n, q, ε in the stated range,*

$$Q(n, q, \varepsilon) \leq Q_{\text{na}}(n, q, \varepsilon) \leq \min \left\{ N, \left\lceil \frac{2^{73} q(q+1) \Gamma_n^q}{\varepsilon} \right\rceil \right\}. \quad (3)$$

The constructed estimator has success probability at least $3/4$. Its random-bit count is deterministically finite. All actual oracle inputs are real product vectors. Only arithmetic, comparisons, exact rank tests, and inverses of known nonsingular matrices are used on unknown data.

The constant is a sufficient proof constant, not a practical sample recommendation. Storage, exhaustive column selection, and interpolation conditioning can be prohibitively expensive. No runtime, bit-complexity, or numerical-stability guarantee is asserted.

The preceding package [3] supplied an all-PSD upper branch with exponential factor asymptotic to $[2\sqrt{3-1/n}]^q$, multiplied by $q(q+1)/\varepsilon$ and a universal constant. Theorem 1.1 lowers that factor and establishes nonadaptivity explicitly. The already established fixed- q characterization

$$Q(n, q, \varepsilon) = \Theta_q(\min\{n^q, \varepsilon^{-1}\})$$

is retained, not claimed as a new conclusion of this round.

Proposition 1.2 (An improvement of the combined upper envelope). *Fix $n = 4$ and take $\varepsilon_q = (2/5)^{q/2}$. The displayed upper bounds in the preceding packages select the 4^q exact-basis branch. The new result gives*

$$Q(4, q, \varepsilon_q) = O \left(q^2 \left(\sqrt{\frac{224}{15}} \right)^q \right), \quad \sqrt{224/15} = 3.86436713 \dots < 4,$$

with a universal implied constant.

Proof. Here $\eta = 4/5$, $b = 8/5$, $a = 56/15$, and $\Gamma_4^2 = 448/75$. Dividing Γ_4 by $\sqrt{2/5}$ gives $\sqrt{224/15}$. The real-sphere branch in [2] has exponential scale 5^q ; its complex-sphere branch is at least a polynomial factor times 4^q ; the ordinary-query simulation and the round-2 hierarchy branch are larger than 4^q . Their explicit constants do not alter the stated comparison for large q . The new sufficient constant is also large: the proposition is an asymptotic query-complexity improvement, not a claim about small instances. $\square$

1.3 Exact recovery in a promised binary-factor class

Let

$$\mathcal{V}_q = (\text{Sym}_2(\mathbb{R}))^{\otimes q} \subseteq \text{Sym}_{2^q}(\mathbb{R}).$$

Equivalently, a matrix in $\mathcal{V}_q$ is unchanged by transposition in any individual local factor. Define

$$\mathcal{C}_{\text{loc}} = \mathcal{V}_q \cap \{M \succeq 0\}, \quad \mathcal{C}_{\text{sep}} = \left\{ \sum_{\ell=1}^L \bigotimes_{j=1}^q A_{\ell j} : L < \infty, A_{\ell j} \in \text{Sym}_2(\mathbb{R}), A_{\ell j} \succeq 0 \right\}.$$

No decomposition or coefficients are supplied to the algorithm.

Theorem 1.3 (Exact binary-factor trace complexity). *For either promise $M \in \mathcal{C}_{\text{sep}}$ or $M \in \mathcal{C}_{\text{loc}}$, the minimum worst-case number of real product queries needed to output the exact trace with probability at least $2/3$ is exactly $q + 1$. The upper bound is deterministic and nonadaptive; the lower bound allows randomized adaptive algorithms.*

This is an exact-estimation theorem for a promised subclass, not a lower bound of $q + 1$ for the relative-error problem. It is proved in Section 7.

1.4 The gap that prevents a complete solution

The blocking lower bound from [2] is

$$Q(n, q, \varepsilon) \geq \frac{1}{80000} \max_{1 \leq r \leq q} \min \left\{ n^r, \frac{\sqrt{\lfloor q/r \rfloor}}{\varepsilon} \right\}. \quad (4)$$

The upper envelope is the minimum of (3) and the retained real-sphere, complex-sphere, ordinary-query, and exact-basis branches. At $n = 2$, $\varepsilon = 1/4$, these still leave

$$\Omega\left(\sqrt{q/\log q}\right) \leq Q(2, q, 1/4) \leq O(q(4/3)^q).$$

The new linear-accuracy branch does not improve that particular upper envelope. No matching general lower bound for (3), and no upper bound matching (4), is proved here. Thus the universal-constant dependence on all three parameters requested by RA-11 remains unresolved in this package.

2 Finite-support probes with exact spherical moments

2.1 A weighted finite distribution

A local formal probe $z \in \mathbb{C}^n$ has the following distribution. With probability $\eta = n/(n+1)$, draw its entries independently from $\{1, -1, i, -i\}$. With the remaining probability $1/(n+1)$, draw an independent uniform coordinate $j \in \{1, \dots, n\}$ and phase $\xi \in \{1, -1, i, -i\}$, and set $z = \sqrt{n} \xi e_j$.

The word “formal” matters only for implementation: store z as (w, k) with $z = \sqrt{k} w$, where $k = 1$ in the phase case and $k = n$ in the coordinate case. Products have an integer squared scale $k = n^r$. No square root is needed in an actual query or in the weighted arithmetic described below.

Lemma 2.1 (Local and product moments). *A local probe satisfies*

$$\mathbb{E}zz^* = I, \quad \mathbb{E}zz^T = 0, \quad \|z\|_2^2 = n$$

and, for every complex matrix F of order n ,

$$\mathbb{E}|z^*Fz|^2 = \eta(\|F\|_F^2 + |\operatorname{tr} F|^2). \quad (5)$$

Consequently

$$\mathbb{E}|z^*Fz - \operatorname{tr} F|^2 = \eta \left(\|F\|_F^2 - \frac{|\operatorname{tr} F|^2}{n} \right). \quad (6)$$

For a product of s independent local probes,

$$\mathbb{E}zz^* = I, \quad \mathbb{E}zz^T = 0, \quad \mathbb{E}|u^*z|^4 \leq \beta^s \|u\|_2^4. \quad (7)$$

Proof. The phase distribution has quadratic second moment $\|F\|_F^2 + |\operatorname{tr} F|^2 - \sum_j |F_{jj}|^2$. The coordinate distribution has moment $n \sum_j |F_{jj}|^2$. Their mixture cancels the diagonal correction because $(1 - \eta)n = \eta$, proving (5). Both components are isotropic, proper, and have squared norm n . Subtract the squared mean, using $1 - \eta = \eta/n$, to obtain (6). Taking $F = uu^*$ gives $\mathbb{E}|u^*z|^4 = 2\eta \|u\|^4$. For the tensor-product bound, condition on all but one factor and apply the local inequality. Cauchy–Schwarz bounds cross products of squared slice contractions; induction gives the factor $(2\eta)^s$. Isotropy and properness tensorize. $\square$

These are exactly the complex-spherical second and fourth moments at local radius $\sqrt{n}$. The distribution is used for these identities; no claim of exhaustive novelty or priority for this elementary moment construction is made. The relevance of partial traces to Kronecker-probe moments is established in [4].

2.2 Complex responses require only real product queries

Lemma 2.2 (Polynomial simulation, retained from round 1). *For any real linear map M and complex factors $z_j = a_j + ib_j$, the response $M(\bigotimes_j z_j)$ is computable using $q + 1$ real product queries.*

Proof. The vector polynomial $p(t) = \bigotimes_{j=1}^q (a_j + tb_j)$ has degree at most q . Query $Mp(t)$ at $t = 0, \dots, q$, and evaluate its Lagrange interpolant at $t = i$:

$$Mp(i) = \sum_{j=0}^q \left(\prod_{\substack{0 \leq h \leq q \\ h \neq j}} \frac{i - h}{j - h} \right) Mp(j).$$

All actual inputs have real factors, and the coefficients are Gaussian rational. Only finitely many arithmetic operations are used. This identity was proved in [2]; it can be badly conditioned in floating-point arithmetic. $\square$

3 Superset-specific controls

3.1 Partial traces and local trace removal

For $S \subseteq [q]$, let $\mathcal{T}_S M = \text{tr}_{S^c} M$ be the unnormalized partial trace retaining the modes in S , in increasing order. Its adjoint-normalized lift is

$$\mathcal{L}_S A = A \otimes I_{S^c} / n^{q-|S|},$$

with factors restored to their original positions. On a retained mode j , let E_j replace that matrix factor by its trace times I/n . The maps E_j are commuting Frobenius-orthogonal projections. For $W \subseteq U$, define on the U -mode matrix space

$$\mathcal{D}_W^{(U)} = \prod_{j \in W} (\text{Id} - E_j).$$

This is an orthogonal projection. Its output has zero partial trace in each mode in W .

Write $A_U = \mathcal{T}_U M$. For $M \succeq 0$, every $A_U \succeq 0$ and $\text{tr} A_U = \tau := \text{tr} M$. Suppose estimates $\hat{A}_U$ are already available for every strict superset $U \supsetneq S$. Define the *separate* control

$$B_S = \sum_{U \supsetneq S} \mathcal{L}_U \mathcal{D}_{U \setminus S}^{(U)} \hat{A}_U, \quad e_U = A_U - \hat{A}_U. \quad (8)$$

The matrices B_S are known. They need not be PSD; they are Hermitian when the estimates are Hermitian.

Lemma 3.1 (Superset decomposition). *For every S ,*

$$\mathcal{T}_S B_S = 0, \quad (9)$$

$$M - B_S = \mathcal{L}_S A_S + \sum_{U \supsetneq S} \mathcal{L}_U \mathcal{D}_{U \setminus S}^{(U)} e_U. \quad (10)$$

The terms in the sum occupy mutually orthogonal blocks of the identity/traceless tensor decomposition.

Proof. Split each local matrix space into scalar multiples of I and traceless matrices. Let P_V project onto tensors whose traceless factors occur exactly at modes V . These projections are mutually orthogonal and sum to the identity. The term $\mathcal{L}_S A_S$ contains exactly the components $V \subseteq S$. For a component $V \not\subseteq S$, the unique set $U = S \cup V$ is a strict superset of S , and

$$P_V \mathcal{L}_U \mathcal{D}_{U \setminus S}^{(U)} A_U = P_V M.$$

It appears in no other term indexed by a strict superset. This proves the same decomposition with e_U replaced by A_U and $B_S = 0$, and subtraction gives (10). Every term of B_S has a traceless factor in some mode outside S , so its partial trace onto S is zero. $\square$

3.2 The exact noise identity

For an outside product probe y on S^c , let $K_y x = x \otimes y$, with factors in their original order, and put $C_y(R) = K_y^* R K_y$.

Theorem 3.2 (Noise depends only on strict supersets). *Freeze the estimates used in (8). Then*

$$\mathbb{E} C_y(M - B_S) = A_S,$$

and

$$\begin{aligned} \mathbb{E} \|C_y(M - B_S) - A_S\|_F^2 \\ = \sum_{U \supseteq S} \eta^{|U|-|S|} \left\| \mathcal{D}_{U \setminus S}^{(U)} e_U \right\|_F^2 \leq \sum_{U \supseteq S} \eta^{|U|-|S|} \|e_U\|_F^2. \end{aligned} \quad (11)$$

For $S = \emptyset$, set $B_* = B_\emptyset$. It has trace zero, and

$$\text{Var}(z^*(M - B_*)z) = \sum_{\emptyset \neq U \subseteq [q]} \eta^{|U|} \left\| \mathcal{D}_U^{(U)} e_U \right\|_F^2 \leq \sum_{U \neq \emptyset} \eta^{|U|} \|e_U\|_F^2. \quad (12)$$

Proof. Compression of $\mathcal{L}_S A_S$ is identically A_S , since every local probe has squared norm n . An error term for U contracts only the modes in $U \setminus S$; all other outside modes contain I/n and contract to one. Each contracted mode has zero partial trace. Applying (5) entrywise therefore multiplies the expected squared Frobenius norm by exactly η at each such mode. The term has mean zero.

For two different supersets U_1, U_2 , choose an outside mode in their symmetric difference. In that mode, one term has zero partial trace and the other has an I/n factor. Condition on all other modes. The cross inner product has expectation zero. Thus the compressed error terms are orthogonal in the L^2 Frobenius space. Summing their squared norms gives the equality in (11); orthogonal projection is a contraction, giving the inequality. The scalar case is exactly (12). $\square$

Remark 3.3 (Difference from the preceding hierarchy). The preceding construction [3] maintained a single shared control and paid for a prefix of previously generated errors. Formula (8) instead builds a distinct control for each target partial trace. Errors indexed by incomparable subsets do not enter its noise bound. This is the principal new step in the all-PSD upper bound.

3.3 A permitted noisy oracle with a PSD mean

At stage S , freeze B_S . On an inside product vector x , draw a_S independent outside probes and return

$$\frac{1}{a_S} \sum_{j=1}^{a_S} K_{y_j}^* (M - B_S) K_{y_j} x = (A_S + F)x. \quad (13)$$

The mean-zero Hermitian matrix F is independent of x . Different calls have independent copies of F , conditional on the frozen history. Its squared Frobenius moment is at most the right side of (11) divided by a_S . The mean is the original PSD matrix A_S , never the possibly indefinite residual. Each outside sample costs one complex full product response, hence $q + 1$ original real calls.

If the inside probe is stored as $x = \sqrt{k} w$ and an outside probe as $y = \sqrt{l} u$, query only the raw product $w \otimes u$. The raw inside response is

$$l K_u^* (M - B_S) K_u w.$$

Averaging these quantities returns $(A_S + F)w$; multiplying it formally by $\sqrt{k}$ gives the response in (13). The scale l is an integer. All contractions use the complete oracle response and known factors; no unstructured input is submitted.

4 Low-rank approximation with weighted sketches

The following lemma is the approximation mechanism proved in round 2. Appendix A reproduces its proof and changes the second-sketch count to a fixed padded size, establishing nonadaptivity explicitly.

Lemma 4.1 (Boosted noisy PSD approximation). *Let $A \succeq 0$ have trace τ , and let formal input vectors be isotropic and proper with fourth-moment bound h . Suppose a call returns $(A + F)x$, where F is mean-zero Hermitian, independent of x , with $\mathbb{E} \|F\|_F^2 \leq \nu$. For an integer $r_0 \geq 1$, a procedure using at most*

$$2^{28} h r_0 \ell_0, \quad \ell_0 = 8q + 17,$$

noisy responses returns a Hermitian $\hat{A}$ such that, except with probability $1/(12K)$,

$$\|\hat{A} - A\|_F^2 \leq 2^{17} (\tau^2 / r_0 + \nu). \quad (14)$$

Inputs and response counts can be chosen independently of the response values. A coordinate basis is used when r_0 is at least the matrix dimension. The selection procedure does not need τ or ν .

4.1 Eliminating formal square roots

For a first sketch, store raw responses y_j with integer squared weights k_j . The formal data matrix has columns $\sqrt{k_j} y_j$. Their column spans are the raw column spans. To select columns, minimize

$$\sum_j k_j \|(I - P)y_j\|^2$$

over the specified finite collection of independent column subsets. This objective and every projector are computable without square roots.

For the independent regression sketch, write its raw inputs and responses as w_j, z_j , with squared scales k_j , and let F_0 be an independent raw-column basis selected from the first sketch. The exact regression formula is

$$\tilde{A} = F_0 \left(F_0^* \left[\sum_j k_j w_j w_j^* \right] F_0 \right)^{-1} F_0^* \left[\sum_j k_j w_j z_j^* \right]. \quad (15)$$

It is the ordinary formal-sketch formula with square-root scales canceled pairwise. If the Gram matrix is singular, return zero. Hermitize the result. Squared Frobenius distances for median selection likewise require no square roots. Known probe factors and interpolation coefficients are Gaussian rational; responses may contain arbitrary real entries from M .

5 Complete schedule and proof of the upper bound

5.1 Budgets for every accuracy

Choose a positive dyadic rational R such that

$$\rho \leq R < 2\rho, \quad \rho = \varepsilon^{-1} (b/a)^{q/2}. \quad (16)$$

Starting with $R = 1$, repeated doubling or halving and comparisons of $R^2 \varepsilon^2 a^q$ with b^q find such a value with finite arithmetic. No square root is computed. For a subset of size s , target the error

$$\|e_S\|_F^2 \leq \delta_s \tau^2, \quad \delta_s = \frac{(3/4)^s}{R}. \quad (17)$$

If $R(4/3)^s \leq 1$, set $\hat{A}_S = 0$ and make no queries at that stage. This is always successful because $A_S \succeq 0$ implies $\|A_S\|_F^2 \leq \tau^2$. Otherwise the stage is active; use

$$r_{0,s} = \left\lceil 2^{18} R (4/3)^s \right\rceil, \quad a_s = \left\lceil 2^{18} b^{q-s} \right\rceil, \quad h_s = \beta^s, \quad \ell_0 = 8q + 17. \quad (18)$$

For the full set use $a_q = 1$, because its noisy oracle is exact. The looser displayed bound remains valid for counting.

Process all nonempty subsets in descending size. At an active stage, construct B_S from strict supersets using (8), supply (13), and apply Lemma 4.1. At an inactive stage store zero. Finally construct B_* and take

$$m = \left\lceil \frac{12(b^q - 1)}{R\varepsilon^2} \right\rceil \quad (19)$$

fresh full product probes. Return

$$\hat{\tau} = \frac{1}{m} \sum_{j=1}^m z_j^* (M - B_*) z_j. \quad (20)$$

The control has trace zero, so the conditional mean is always τ . Each summand is real. Inactive stages are essential: they avoid an exponential ceiling overhead when the balancing value R is small. There is no separate large-error branch.

5.2 Probability proof under the ideal finite distribution

Condition on a fixed nonzero $M \succeq 0$ and on all strict-superset stages satisfying (17). At an active S of size s , Theorem 3.2 bounds the unaveraged noise moment by

$$\begin{aligned} \tau^2 \sum_{U \supsetneq S} \eta^{|U|-s} \delta_{|U|} &= \delta_s \tau^2 \sum_{j=1}^{q-s} \binom{q-s}{j} (3\eta/4)^j \\ &= \delta_s \tau^2 (b^{q-s} - 1). \end{aligned} \quad (21)$$

After averaging, $\nu \leq \delta_s \tau^2 / 2^{18}$, and $\tau^2 / r_{0,s} \leq \delta_s \tau^2 / 2^{18}$. Therefore (14) gives (17), except with conditional probability $1/(12K)$. Inactive stages succeed deterministically. A first-failure union bound makes every stage successful with probability at least $11/12$; independence across stages is not assumed.

On a successful hierarchy, the final variance is at most

$$\tau^2 \sum_{S \neq \emptyset} \eta^{|S|} \delta_{|S|} = \frac{(b^q - 1)\tau^2}{R}.$$

The independent average in (20) therefore has conditional variance at most $\varepsilon^2 \tau^2 / 12$. Chebyshev gives conditional failure at most $1/12$. Total failure under ideal finite sampling is at most $1/6$. The zero input produces zero responses and estimates throughout, and is handled exactly.

5.3 Counting actual oracle calls

At an active stage, $R(4/3)^s > 1$, hence

$$r_{0,s} \leq 2^{19} R(4/3)^s, \quad a_s \leq 2^{19} b^{q-s}.$$

The total number of complex calls for all approximation stages is at most

$$\begin{aligned} 2^{66} \ell_0 R \sum_{s=1}^q \binom{q}{s} \left(\frac{4\beta}{3} \right)^s b^{q-s} &\leq 2^{66} \ell_0 R \left(b + \frac{8\eta}{3} \right)^q \\ &= 2^{66} \ell_0 R a^q < \frac{2^{67} \ell_0 \Gamma_n^q}{\varepsilon}. \end{aligned}$$

The final sample count is at most $12\Gamma_n^q/\varepsilon + 1 \leq 13\Gamma_n^q/\varepsilon$. Since $\ell_0 \leq 17q < 32q$, the total complex count is at most

$$2^{73}q\Gamma_n^q/\varepsilon.$$

Every complex response uses $q + 1$ original real calls. The deterministic N -call coordinate-basis algorithm is compared with the explicit hierarchy budget before any calls, and the cheaper branch is selected. This proves the call bound in Theorem 1.1.

5.4 Finite-bit sampling and the success probability

The ideal local law can be sampled by a uniform integer in an alphabet of size $n + 1$, and, only on its coordinate branch, a uniform integer in an alphabet of size n . Phases use two fair bits each. Uniform integers whose alphabet size is not a power of two ordinarily use rejection sampling; an unbounded loop is not necessary here.

Let T_C be the explicit deterministic complex-call budget of the chosen hierarchy. At most $2qT_C$ finite-uniform integer draws are needed: inside probes can be reused across an outside average, and $a_s \geq 1$. Set

$$J = \lceil \log_2(24qT_C) \rceil.$$

For an alphabet of size d , draw $\lceil \log_2 d \rceil$ bits, accept an integer below d , and allow at most J attempts. If all attempts fail, return zero from the algorithm. Each draw fails with probability at most 2^{-J} . Conditional on acceptance its value is exactly uniform.

Couple this bounded sampler to the ideal rejection sampler. The probability that any used draw is truncated is at most $2qT_C2^{-J} \leq 1/12$. Thus success is at least $5/6 - 1/12 = 3/4$. Random-bit use is bounded deterministically by a finite function of n, q, ε . The reference records sampling aborts distinctly; it never silently substitutes a biased integer draw.

5.5 Why the algorithm is nonadaptive

Rank targets, active stages, averaging counts, and candidate counts depend only on n, q, ε . Both random sketch input lists can be selected in advance. In Appendix A, the regression sketch is padded to a count based on $2r_0$, not on the observed basis rank; its calls are still issued when the basis is empty. A coordinate sweep is chosen from the known comparison $r_0 \geq n^s$.

The controls, selected bases, Gram inverses, and candidate medians affect only postprocessing. Every raw full input is a product of an independently preselected inside probe and outside probe. These remain unchanged by the control subtraction. The finite-sampling abort can also be determined before any matrix responses are requested. Consequently the complete query list is nonadaptive. This completes the proof of Theorem 1.1.

6 A tunable geometric family

The factor $3/4$ is chosen for a compact rational schedule. More generally, fix any positive rational t . Use error targets

$$\delta_s = (t/\eta)^s/R, \quad b_t = 1 + t, \quad a_t = 1 + t + 2\eta^2/t.$$

The same proof, with zero approximations when $\delta_s \geq 1$, gives

$$Q(n, q, \varepsilon) \leq O\left(\frac{q(q+1)}{\varepsilon} [(1+t)(1+t+2\eta^2/t)]^{q/2}\right), \quad (22)$$

with a universal sufficient constant independent of the rational choice of t . Indeed the strict-superset noise sum is $\delta_s \tau^2 [(1+t)^{q-s} - 1]$, while the learning sum has base $1 + t + 2\eta^2/t$.

Differentiating the squared exponential factor gives a unique positive minimizer t_* satisfying

$$t_*^2(1 + t_*) = \eta^2.$$

At this value the factor is $(1 + t_*)\sqrt{1 + 2t_*}$. Rational approximations may be used in (22); no irrational constant need be evaluated on unknown data. The displayed main theorem instead takes $t = 3\eta/4$, which yields the explicit rational square Γ_n^2 . No optimality of this algorithm family, or of its exponential factor for RA-11, is claimed.

7 Exact trace for real-separable binary factors

7.1 The deterministic upper bound

Put $z_0 = e_1 + ie_2$ and $z = z_0^{\otimes q}$. For every real symmetric matrix A of order two,

$$z_0^* A z_0 = A_{11} + A_{22} = \text{tr } A.$$

Tensoring and taking real linear combinations gives

$$z^* M z = \text{tr } M \quad (M \in \mathcal{V}_q). \quad (23)$$

By Lemma 2.2, Mz is obtained from $q + 1$ real product calls. This proves the upper bound in Theorem 1.3; PSD is not needed for this identity. The interpolation device is inherited from [2]; its application to this larger cone and the matching lower bound are the conclusions here.

The promise is essential. For

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad M = I_4 + \frac{1}{2}J \otimes J,$$

M is real PSD, with eigenvalues $1/2$ and $3/2$. But $z^* M z = 2$ whereas $\text{tr } M = 4$. Thus (23) is not an identity for arbitrary real PSD matrices.

7.2 A product projector in every short-query kernel

Lemma 7.1 (Annihilating projector). *For any $t \leq q$ real product inputs $v^{(1)}, \dots, v^{(t)}$ in $(\mathbb{R}^2)^{\otimes q}$, there is a projector $P \in \mathcal{C}_{\text{sep}}$ of rank 2^{q-t} with $Pv^{(j)} = 0$ for all j .*

Proof. Assign query j to local mode j . Choose a unit $x_j \in \mathbb{R}^2$ perpendicular to that query's factor in mode j ; choose any unit vector if the factor is zero. Set

$$P = \bigotimes_{j=1}^t x_j x_j^T \otimes I_2^{\otimes (q-t)}.$$

Its contraction with query j vanishes in the assigned factor. It is a real product of PSD matrices, is a projector, and has the stated rank. Exact normalization can instead use $xx^T/(x^T x)$, requiring no square root. $\square$

A seed-dependent zero-transcript adversary alone would not prove a randomized lower bound. The next argument supplies a single input prior independent of the seed.

7.3 The adaptive Gaussian argument

Equip $\mathcal{V}_q$ with the Frobenius inner product and choose an orthonormal real basis. Let G have independent standard Gaussian coefficients in that basis. For a fixed algorithmic seed, condition on the complete transcript of at most $t \leq q$ adaptive full-vector product queries.

Lemma 7.2 (Conditional trace variance). *Under this unconditioned Gaussian prior, the conditional trace is a nondegenerate Gaussian. More quantitatively,*

$$\text{Var}(\text{tr } G \mid \text{transcript}) \geq 2^{q-t}.$$

Proof. A query response consists of linear functionals of G . Initially the covariance operator on $\mathcal{V}_q$ is the identity. Inductively, conditional on the past, the next query is fixed and the remaining input law is Gaussian on an affine subspace. Conditioning on its linear response leaves covariance equal to the orthogonal projection onto the intersection of the previous kernel and the new response kernel. Thus after all calls the covariance is the projector onto

$$\mathcal{K} = \{H \in \mathcal{V}_q : Hv^{(j)} = 0 \text{ for every queried } j\}.$$

This induction, rather than conditioning on an adaptive span as though it were independent of the input, handles adaptivity.

The trace functional has representing vector I . Lemma 7.1 supplies a $P \in \mathcal{K}$ with $\text{tr } P = \|P\|_F^2 = 2^{q-t}$. Cauchy–Schwarz therefore gives

$$\|\text{proj}_{\mathcal{K}} I\|_F^2 \geq \frac{\langle I, P \rangle^2}{\|P\|_F^2} = 2^{q-t}.$$

This is precisely the conditional trace variance. Linear dependence among queries does not invalidate the argument. A shorter execution may be padded by zero queries. $\square$

For any transcript-based output, exact equality with a nondegenerate conditional Gaussian has probability zero. This remains true after averaging over the algorithm’s independent seed. It remains to make the prior supported on promised PSD inputs, without losing this zero-success assertion.

Lemma 7.3 (A real-separable neighborhood). *The cone $\mathcal{C}_{\text{sep}}$ contains a nonempty relatively open subset of $\mathcal{V}_q$ consisting of positive-definite matrices.*

Proof. The four local rank-one projectors

$$P_1 = e_1 e_1^T, \quad P_2 = e_2 e_2^T, \quad P_3 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad P_4 = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

span $\text{Sym}_2(\mathbb{R})$ and sum to $2I_2$. Their 4^q tensor products span $\mathcal{V}_q$ and sum to $2^q I$. The linear map taking their coefficient vector to the corresponding matrix is surjective. An open neighborhood of the strictly positive coefficient vector with every entry 2^{-q} therefore maps onto a neighborhood of I in $\mathcal{V}_q$. This image lies in the real-separable cone. Intersect with a sufficiently small ball around I to ensure positive definiteness. $\square$

Proof of the lower bound in Theorem 1.3. Let $\mathcal{O}$ be the open set from Lemma 7.3. The Gaussian prior has positive probability of belonging to $\mathcal{O}$. The unconditional probability of exact success with at most q calls is zero by Lemma 7.2; its intersection with $\{G \in \mathcal{O}\}$ also has probability zero. Consequently the prior conditioned on $\mathcal{O}$ is supported on real-separable positive-definite matrices

and still gives average success zero. Uniform success at least $2/3$ on that class is impossible. This also rules out such an algorithm on the larger $\mathcal{C}_{\text{loc}}$.

An algorithm originally defined only for promised PSD inputs may be extended measurably on other transcripts for this argument, with arbitrary zero outputs on undefined branches. Its behavior on $\mathcal{O}$ is unchanged. Finite exact-arithmetic algorithms give measurable query rules; no extra computation by the lower-bound prior is assumed. Together with the deterministic upper bound, this proves the exact rate $q + 1$. $\square$

8 Source audit, computation, and limitations

The underlying ordinary-query variance-reduction ideas are prior work, including Hutch++ [6]; the noisy two-sided approximation lemma used here was developed and checked in the preceding package [3]. The new all-PSD mechanism is the superset-specific decomposition and its geometric schedule. The complex-query simulation comes from round 1 [2]. Neither a conditioned scalar-query lower bound from [5] nor an estimator-specific variance lower bound from [4] is promoted to an unrestricted full-vector minimax lower bound.

The original two packages are included unchanged in the archive directory. Their source claims and numerical evidence are not relabeled as new results. The present note is not independently reviewed, and no formal theorem prover has certified it. A separate proof-scope audit identifies each oracle, positivity, conditioning, and randomness requirement.

The exact checks cover finite-support moments, the deterministic superset decomposition, exhaustive contraction-noise identities, the geometric schedules, finite-uniform sampler truncation, exact weighted regression, the binary-factor trace identity, and its explicit failure outside the promise. A certified basis-branch execution and a preflight rejection before any calls are recorded. A small-constant full hierarchy is also executed in exact arithmetic; it is explicitly diagnostic.

The numerical suite uses floating-point linear algebra and reduced constants. It tests complete real-input oracle executions, rather than silently substituting an unstructured matrix-vector oracle. These trials test implementation behavior, not the universal sufficient schedule. The certified hierarchy at its enormous sufficient constants is not executed. Arithmetic cost, memory cost, and stability are not covered by the query theorem.

From the package root, run `bash run_checks.sh` to regenerate evidence and `bash build.sh` to compile this note. Exact per-test counts, configurations, and outputs are in the JSON files in `results/`. The PDF is rendered and visually checked separately from mathematical verification. The final status remains **partial**: the bounds in Section 1.4 do not match.

A Proof of the noisy approximation lemma

This appendix reproduces the round-2 argument [3], with the fixed-size regression padding noted explicitly. Its assumptions concern formal isotropic inputs, so the weighted representation in Section 4 implements it without changing the proof.

A.1 A lower covariance estimate

Lemma A.1 (Elementary covariance injection). *Suppose $g \in \mathbb{C}^r$ satisfies $\mathbb{E}gg^* = I$, $\mathbb{E}gg^T = 0$, and $\mathbb{E}|u^*g|^4 \leq h\|u\|^4$. For $m \geq 2^{25}hr$ independent copies,*

$$\mathbb{P} \left\{ \frac{1}{m} \sum_{j=1}^m g_j g_j^* \succeq \frac{1}{4} I \right\} \geq 63/64.$$

Proof. First consider an isotropic real vector $w \in \mathbb{R}^r$ with fourth-moment bound h . For a unit u , put $\phi(u^T w) = \min\{(u^T w)^2, 4h\}$. Then $\mathbb{E}\phi(u^T w) \geq 3/4$, since

$$\mathbb{E}[(u^T w)^2 \mathbf{1}_{(u^T w)^2 > 4h}] \leq \frac{\mathbb{E}(u^T w)^4}{4h} \leq 1/4.$$

The scalar function ϕ is $4\sqrt{h}$ -Lipschitz and vanishes at zero. Symmetrization and the scalar Rademacher contraction inequality give

$$\begin{aligned} \mathbb{E} \sup_{\|u\|=1} \left| \frac{1}{m} \sum_j \phi(u^T w_j) - \mathbb{E}\phi(u^T w) \right| &\leq 16\sqrt{h} \mathbb{E} \left\| \frac{1}{m} \sum_j \epsilon_j w_j \right\|_2 \\ &\leq 16\sqrt{hr/m}. \end{aligned}$$

The first step uses symmetrization factor two and contraction factor two for the absolute supremum; the second uses independence of the Rademacher signs and $\mathbb{E}\|w\|^2 = r$. These inequalities apply to finite nets and then, by continuity and domination by a bounded function, to the unit sphere. For $m \geq 2^{22}hr$, Markov's inequality makes the displayed supremum at most $1/2$ except with probability $1/64$. Hence every empirical clipped second moment is at least $1/4$, and the unclipped second moment is larger.

For the complex claim, realify g as $w = \sqrt{2}(\Re g, \Im g) \in \mathbb{R}^{2r}$. Properness makes this vector isotropic, and its real fourth-moment bound is at most $4h$. The real result needs $2^{22}(4h)(2r) = 2^{25}hr$ samples. Apply its Gram inequality to the real and imaginary linear forms of each u^*g and add; the complex Gram bound follows. $\square$

Fourth-moment lower-covariance bounds are established tools; see, for example, Oliveira [7]. The deliberately loose constant above is derived here so that no unquantified subspace-embedding assertion is needed.

A.2 Column selection without exact singular vectors

Lemma A.2 (Finite column selection). *For any complex matrix Y and integer $r_0 \geq 1$, one can select at most $2r_0$ columns whose orthogonal projector P satisfies*

$$\|(I - P)Y\|_F^2 \leq 2\|Y - Y_{r_0}\|_F^2. \quad (24)$$

Selection can be done by finite arithmetic, exact rank tests, and exhaustive comparison of projection residuals; no new oracle calls or exact SVD are required.

Proof. If $\text{rank } Y \leq 2r_0$, select a basis of its columns. Otherwise put $k = 2r_0$. For every independent k -column submatrix Y_J , assign weight $w_J = \det(Y_J^* Y_J)$ and let P_J project onto its span. A Gram determinant Schur-complement identity gives

$$\sum_{|J|=k} w_J \|(I - P_J)Y\|_F^2 = (k+1)e_{k+1}, \quad \sum_{|J|=k} w_J = e_k,$$

where e_j is the j th elementary symmetric polynomial of the squared singular values $\lambda_1 \geq \lambda_2 \geq \dots$ of Y . Indeed each $(k+1)$ -column determinant is counted once for each of its $k+1$ omitted columns in the first sum. Cauchy–Binet identifies the sums of principal minors with e_j .

Every $(k+1)$ -element index set contains at least $k - r_0 + 1$ indices larger than r_0 . Counting their contributions to $e_k \sum_{j>r_0} \lambda_j$ gives

$$(k - r_0 + 1)e_{k+1} \leq e_k \sum_{j>r_0} \lambda_j.$$

Therefore the weighted mean projection error is at most $(k+1)/(k-r_0+1)$ times the best rank- r_0 squared residual, and this ratio is less than two. At least one subset is no worse than its weighted mean. Exhaustive minimization over the finitely many independent subsets finds one. Every residual is computable from $Y_J(Y_J^*Y_J)^{-1}Y_J^*$. $\square$

The column-count/error tradeoff is a standard column-subset result [8]; the argument above also supplies its complex version and the finite-arithmetic implementation used here. The search may be prohibitively expensive, but arithmetic cost is not charged in RA-11.

A.3 Two-sided noisy reconstruction

Lemma A.3 (Noisy PSD approximation). *Let $A \succeq 0$ be Hermitian of order d , with trace τ . Suppose inputs can be drawn from an isotropic proper complex distribution with fourth bound h , and a noisy call on x returns $(A+F)x$, where F is an independent copy of a fixed mean-zero Hermitian random matrix, independent of x , with $\mathbb{E}\|F\|_F^2 \leq \nu$. For each integer $r_0 \geq 1$, a finite-arithmetic algorithm using at most $2^{28}hr_0$ noisy calls returns a Hermitian $\hat{A}$ such that*

$$\mathbb{P}\left\{\|\hat{A} - A\|_F^2 \leq 2^{13}(\tau^2/r_0 + \nu)\right\} \geq 7/8. \quad (25)$$

A coordinate basis may be used instead of random inputs when $r_0 \geq d$.

Proof. If $r_0 \geq d$, query a basis and Hermitize the resulting noisy matrix. Identically distributed noise matrices give expected squared error $\sum_j \mathbb{E}\|F_j e_j\|^2 = \mathbb{E}\|F\|_F^2 \leq \nu$; Markov proves the stated bound.

Now suppose $r_0 < d$. Put $J_0 = 2^{25}$ and $m = \lceil J_0 hr_0 \rceil$. For analysis, normalize all sketch columns by $1/\sqrt{m}$ and write

$$Y = A\Omega + E.$$

Select the at most $2r_0$ columns provided by Lemma A.2, and let P be their orthogonal projector. An orthonormal basis U for its range is used only in the proof. Put $k = \text{rank } P \leq 2r_0$.

Diagonalize A for analysis as $A = U_1\Lambda_1U_1^* + U_2\Lambda_2U_2^*$, splitting after r_0 eigenvalues. Set $A_2 = U_2\Lambda_2$ and $\Omega_j = U_j^*\Omega$, and put $t_0 = \|A - A_{r_0}\|_F^2 \leq \tau^2/r_0$. With probability at least $63/64$, the top block Ω_1 has $\|\Omega_1^\dagger\|_2 \leq 2$, by Lemma A.1. Also

$$\mathbb{E}\|A_2\Omega_2\|_F^2 = t_0, \quad \mathbb{E}\|E\|_F^2 \leq \nu.$$

Thus each is at most 64 times its mean bound except on an event of probability $1/64$ (a zero mean bound gives zero almost surely).

The column selection guarantee implies

$$\|(I-P)Y\|_F^2 \leq 2\|Y - Y_{r_0}\|_F^2 \leq 4(\|A_2\Omega_2\|_F^2 + \|E\|_F^2).$$

Reconstructing the leading block through $\Omega_1^\dagger$, and bounding each sum of two matrices by twice the sum of squared norms, gives

$$\begin{aligned} \|(I-P)A\|_F^2 &\leq t_0 + 48(\|A_2\Omega_2\|_F^2 + \|E\|_F^2) \\ &\leq 3073t_0 + 3072\nu. \end{aligned} \quad (26)$$

For clarity, the first factor 48 comes from $8\|(I-P)Y\|_F^2 + 8\|A_2\Omega_2 + E\|_F^2$ for the leading-block error, followed by the preceding display. No independence between spectral blocks of Ω has been assumed.

Take an independent normalized sketch Ψ with the fixed count $m' = \lceil 2J_0hr_0 \rceil$ columns and observe $Z = A\Psi + F'$. This count is chosen before the first sketch is observed. Its calls are still issued when $k = 0$; in that case return zero afterward, as (26) already suffices. When $k > 0$, put

$$G = U^*\Psi, \quad \tilde{A} = U(GG^*)^{-1}GZ^*.$$

If the Gram matrix is singular, return zero on that branch. Conditional on all first-stage data, it is nonsingular and has inverse norm at most four except with probability $1/64$. Let $R_0 = (I - P)A$. Since $U^*R_0 = 0$, independent columns and fourth-moment Cauchy–Schwarz give

$$\mathbb{E} \|U^*\Psi\Psi^*R_0\|_F^2 \leq \frac{hk}{m'} \|R_0\|_F^2. \quad (27)$$

Here each entry is an average of centered products; for vectors u, v , $\mathbb{E}|u^*x|^2|v^*x|^2 \leq h \|u\|^2 \|v\|^2$. Also $\mathbb{E} \|F'\|_F^2 \leq \nu$. Markov again bounds both squared norms by 64 times their respective right sides. The residual orthogonal to U and the fitted coefficient error are Frobenius orthogonal, so

$$\begin{aligned} \|\tilde{A} - A\|_F^2 &\leq \left(1 + \frac{2048hk}{m'}\right) \|R_0\|_F^2 + 512\nu \\ &\leq 6146t_0 + 6656\nu \leq 2^{13}(\tau^2/r_0 + \nu). \end{aligned}$$

The noise term uses $\|(GG^*)^{-1}G\|_2^2 \leq 4$; it does not require independence between the noise columns and their input vectors. Hermitization $(\tilde{A} + \tilde{A}^*)/2$ cannot increase error relative to Hermitian A . The six exceptional events total at most $6/64 < 1/8$. Finally, $m \leq 2J_0hr_0$, $m' \leq 4J_0hr_0$, and $m + m' \leq 6J_0hr_0 < 2^{28}hr_0$. $\square$

Remark A.4 (Actual arithmetic and actual inputs). An arbitrary independent-column basis F_0 of the selected columns can replace U . The exact formula is

$$\tilde{A} = F_0(F_0^*\Psi\Psi^*F_0)^{-1}F_0^*\Psi Z^*.$$

Using unnormalized raw sketches gives the same result, since normalization factors cancel. Rank tests, inverses, projections, and comparisons of squared residuals use only finitely many real arithmetic operations and comparisons. No spectral decomposition or square root is performed by the exact algorithm. The random inputs are product probes, and the basis-sweep inputs are coordinate products. In particular, the algorithm never requests AU or AF_0 .

A.4 Boosting without the unknown trace

Lemma A.5 (Metric-median boosting). *Run Lemma A.3 independently $\ell_0 = 8q + 17$ times, conditional on the frozen stage history. Among the candidates, select one whose median squared Frobenius distance to the others (including itself) is smallest. Then, except with probability at most $1/(12K)$,*

$$\|\hat{A} - A\|_F^2 \leq 2^{17}(\tau^2/r_0 + \nu). \quad (28)$$

Selection needs neither τ nor a known good-error radius.

Proof. If a strict majority lies in the radius- σ ball around A , a good candidate has median distance at most 2σ . The selected candidate therefore has at least half its neighbors within 2σ , including some good candidate, so it lies within 3σ of A . The squared bound is multiplied by at most nine, and $9 \cdot 2^{13} < 2^{17}$. Each independent failure probability is at most $1/8$. Hoeffding's inequality gives probability at most $e^{-9\ell_0/32}$ for half or more failures. Since $9\ell_0/32 \geq 3 + q > \log(12K)$ for $q \geq 2$, this is at most $1/(12K)$. $\square$

B Retained general bounds and provenance

For reference, the first package’s integer upper bounds are

$$\begin{aligned} U_0 &= n^q, \\ U_R &= \left\lceil 3[(3n/(n+2))^q - 1]/\varepsilon^2 \right\rceil, \\ U_C &= (q+1) \left\lceil 3[(2n/(n+1))^q - 1]/\varepsilon^2 \right\rceil, \\ U_{\text{ord}} &= n^{q-1}(5\lceil 4/\varepsilon \rceil + 4). \end{aligned}$$

The round-2 branch is

$$U_2 = \left\lceil 2^{71} q(q+1)(2^q - 1) \left\lceil \sqrt{(3 - 1/n)^q - 1} \right\rceil / \varepsilon \right\rceil.$$

The new branch may be added to this minimum; a prior valid branch need not be discarded. The lower bound (4), the high-accuracy rate $\Theta(n^q)$ for $\varepsilon \leq n^{-q}$, and the tensor-product-matrix rate $\Theta(\min\{n, \sqrt{q}/\varepsilon\})$ are retained from [2]. They are not independently re-proved or strengthened by this appendix. The original manuscripts and code remain available in the included unmodified archives.

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